

Karthik Manda

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Education

Nanyang Technological University Singapore

Aug. 2024 – Present

Bachelor of Engineering (Hons), Computer Engineering

- **Relevant Coursework:** Data Structures & Algorithms, Object Oriented Programming, Operating Systems, C/C++, Microprocessor System Design, Embedded Programming.

Machine Learning Specialization – DeepLearning.AI / Stanford Online

2026

- Supervised & unsupervised learning, neural networks, ML engineering best practices; completed full Andrew Ng curriculum.

Experience

WorldQuant BRAIN Remote

Alpha Researcher — Gold Tier (Top 5% Globally)

2026 – Present

- Engineered robust, data-driven alpha factors using financial signals and operational income parameters, applying cross-sectional normalisation to construct low-correlation alphas.
- Achieved Gold tier through sustained alpha performance; Sharpe-optimised factor design directly informs quantitative trading system architecture.

Wee Kim Wee School of Comm. & Info, NTU Singapore

Feb. 2026 – Mar. 2026

Web Development Intern

- Architected and deployed a full-stack Conspiracy Theory Research Datasite in React/JavaScript, delivering 4 production pages with layered content hierarchy across the complete SDLC from requirements to release.
- Engineered data visualisation components and an intuitive CMS workflow, achieving WCAG 2.1 AA compliance throughout.

Projects

Multi-Agent RL Trading System — Python, PyTorch, MADDPG, Pandas

2026 – Present

- Designed a PyTorch-based MADDPG system to autonomously execute coordinated trading decisions across 50+ assets, outperforming single-agent baselines with a Sharpe ratio of 1.5–2.0.
- Engineered OHLCV-based feature pipelines with portfolio-level reward shaping.

STM32 Gimbal Self-Levelling System — Bare-Metal C, STM32, FreeRTOS, I2C

Jan. 2026 – Present

- Engineered a self-levelling gimbal using PID control with Kalman and complementary filtering for IMU sensor fusion.
- Configured TIM, RCC, and GPIO registers directly for PWM motor control, avoiding HAL abstraction entirely.

CodeRecon: Modular Static Analysis Engine — Python, AST, Ollama

Jan. 2026 – Present

- Built a repository-scale AST parser scanning any codebase; detects structural anti-patterns across refactor categories including dead code, complexity hotspots, and coupling violations.
- Local-first LLM inference pipeline generates architecture-level refactor suggestions with zero-cloud data exposure.

Awards & Hackathons

Apple Swift Student Challenge 2026 Global — Winner, Global Top 300

Feb. 2026 – Mar. 2026

- Selected for ArchLab, a system design simulator leveraging Swift 6 concurrency and real-time simulation logic.

HTX HackX 2025 Singapore — Finalist, Top 15 out of 60 Teams

Sep. 2025 – Nov. 2025

- Developed an incident-command dashboard for SCDF hazardous chemical plume response, translating predictive crowd geolocation datasets into actionable operational visualisations.

Professional Development

J.P. Morgan Forage Virtual Experience Programs

2026

- **Software Engineering:** Resolved repository bugs and visualised live trade data using J.P. Morgan's open-source Perspective framework.
- **Investment Banking:** Built DCF and comparable company models covering WACC, terminal value, and sensitivity analysis in an M&A advisory context.
- **Quantitative Research:** Modelled loan default probabilities; developed a FICO score bucketing model and prototype credit risk pricing framework.

AMD AI Developer Programme Virtual Experience

2025

- **AI Application Development & LLM Fine-Tuning:** Built and evaluated AI-powered applications on AMD's open-source GPU ecosystem (ROCm); gained practical deployment pipeline exposure outside NVIDIA-centric toolchains.

Technical Skills

Languages	Python, C/C++, Java, JavaScript/TypeScript (React, Tailwind), Swift, Verilog, SQL.
Hardware	STM32, MSP432, Raspberry Pi, FreeRTOS, I2C/SPI/UART, CAN Bus, TCP/IP.
ML/AI & Quant	PyTorch, Deep RL, LLM Fine-Tuning, Context Engineering, MCP, scikit-learn, Pandas, TensorFlow, NumPy.
Dev Tools	Git, CI/CD (GitHub Actions), Docker, PostgreSQL, Supabase, Firebase, Vivado, Figma.